Maintenance Hub

Description:

This project is a small desktop application designed for technicians. The goal is to make the repair process easier, faster, and more organised. Device details can be entered and choose the type of service, track job status, and calculate repair costs. The system helps the shop work more professionally and reduces errors.

The application uses object-oriented programming. There is a main class called Device, and six device types: Laptop, Phone, Tablet, Desktop, Smartwatch, and OtherDevice. Each device type has different abilities. For example, laptops and desktops support repairs, software services, and data recovery. Phones and tablets support repairs and software services. Smartwatches and other devices only support basic repairs. This is done using interfaces such as Repairable, SoftwareService, and Recoverable.

The system uses three abstract data types (ADTs):

- A Queue to store devices waiting for service
- A LinkedList to store all repair jobs (active and completed)
- A Stack to store undo actions

Each repair job is represented by a RepairJob object. It contains the device, the service type, and the job status. The service types include Diagnostic, Repair, Maintenance, Data Recovery, and Software Service. The job status can be Pending, In Progress, Completed, On Hold, or Cancelled.

The cost of a job is not stored directly. Instead, each device type has its own calcCost() method. When the user wants to see the cost, the system calls this method. This makes the design flexible and easy to update later.

The goal of this application is to support a tech repair shop by:

- Keeping all repair jobs in one place
- Tracking devices from the moment they arrive until the job is completed
- Helping staff choose the correct service type
- Showing job status clearly
- Calculating repair costs automatically
- Reducing paperwork and human error

Conclusion:

The system is small but practical. It can be used by a single technician or with a few updates could be integrated into a more scalable system, where it would function as hub where users can get in touch with technicians to solves their issues.

Diagram: